FMA Sub-Air v1.7.1 — User Manual

FMA Series · Free Mix Audio

License notice (summary). © 2026 Free Mix Audio. All rights reserved.

This plugin is authorized for use as-is only: do not modify its code, and do not redistribute or share it without prior notice and written authorization from Free Mix Audio. The full binding terms are in the `LICENSE.txt` file included in this package.

Version note: v1.7.1 is sonically identical to v1.7.0; only the file header changed (no changelog, license notice added).

---

1. What is FMA Sub-Air

FMA Sub-Air is a sub and air sculptor for mix bus and mastering, with a built-in Vintage saturator. In a single insert it solves the four problems that show up most at the end of a mix:

1. **A weightless low end**: thin, bodiless, and turning muddy the moment you try to lift it with EQ.
1. **A low end with no punch**: the kick and bass are there, but they don't hit.
1. **A low end that collapses in mono**: on a phone, in a club or on any mono system, the sub vanishes because the lows arrive out of phase.
1. **An air that bites**: opening the top end makes the mix harsh and fatiguing.

Sub-Air gives each problem its own tool, under one golden rule: **it never wrecks the mix**. Every section has guardrails that stop an enthusiastic move from destroying previous work. It is an original design, built from scratch for this job — not a copy of any hardware unit or any known plugin.

2. Philosophy: power with guardrails

Most low-end processors are either too shy or too dangerous. Sub-Air is built so you can push with confidence:

- **The sub boost only weighs the sub band.** It is not EQ across the whole mix: the plugin splits the sub band with a 4th-order Linkwitz-Riley

crossover and works on that band alone. Everything else stays untouched.

- **Punch only adds.** TIGHT enhances attacks; it never cuts or compresses.
- **Air has a policeman.** CLEAN watches the high boost and trims only the boost when it detects harshness. Your source is never touched.
- **The limiter is last.** A fixed −1 dBFS ceiling protects the output without coloring anything when it isn't needed.
- **LOW CUT cleans before the limiter.** It removes infrasonics and DC so the limiter works on music only, not on inaudible garbage.

With the SUB controls at zero, the sub section is bit-transparent: the crossover disconnects entirely. With LOW CUT at OFF, the bypass is exact. With DRIVE at 0 %, there is no saturation at all. What you don't use, doesn't sound.

3. Signal flow (the real order)

Audio travels through the plugin in this order. Knowing it explains why each control behaves the way it does:

```
INPUT → [SUB: LR4 crossover → SUB ENHANCE → TIGHT → MONO BELOW] → sum
      → [AIR: 8 kHz shelf → CLEAN (dynamic de-harsh)]
      → [DRIVE: FOCUS → Vintage saturation ×2]
      → [LOW CUT] → [SAFETY LIMITER −1 dBFS] → OUTPUT
```

- **SUB** comes first because weight and punch are decided in the low band, not afterwards.
- **AIR** comes after the sum, so the brightness is applied to the mix once it already has weight.
- **DRIVE** comes after air, so the saturation has something to bite on across the spectrum (or only in the band you pick with FOCUS).
- **LOW CUT** sits at the end, after the sum and before the limiter: this way it also cleans up infrasonics and DC the DRIVE itself may generate.
- **SAFETY LIMITER** is absolutely last: nothing after it can exceed −1 dBFS.

4. Two instruments in one: BUS and INST

The OPERATION switch changes the whole character of the plugin. It is not a preset: it changes the crossover, the de-harsh sensitivity and the drive voicing.

	BUS (left)	**INST (right)**
Intended use	Mix bus, mastering	Individual tracks and instruments

	BUS (left)	INST (right)
Sub crossover	160 Hz	90 Hz
De-harsh threshold	−24 dBFS	−32 dBFS (more sensitive)
Drive voicing	Standard (1.6 / 0.28)	Warmer, stronger even harmonics (1.3 / 0.35)
CLEAN ceiling	Maximum cut ≈ 3 dB	Maximum cut ≈ 5.3 dB

Rule of thumb: processing a full mix, use BUS. Processing a kick, a bass or a single instrument, use INST. The DE-HARSH LED will tell you if the chosen mode is working hard: in INST it responds to normal material, and that is normal — it is a creative tool, not an alarm.

5. Controls, one by one

INPUT GAIN (−12 to +12 dB, 0.1 steps, default 0)

The input trim. It is not just a volume: it sets how hard the signal hits the SUB section and the DRIVE. If the input VU hovers around 0 on strong hits, you are in the zone. Too low in, and TIGHT and DRIVE respond weakly; too hot in, and the limiter works overtime.

SUB BYPASS (lever, default: section active)

Disconnects the entire SUB section (crossover, enhance, punch, mono) in one move. Use it to compare: with the section out, did the low end really get better? The honest comparison is the one that rules.

SUB ENHANCE (0–100 %, default 0)

The weight of the low end. Technically a one-pole shelf at 55 Hz, from 0 to +7.5 dB, applied **only to the sub band** extracted by the crossover. That is why you can push it without muddying the mids: you are not EQ'ing the mix, you are weighing its foundation.

How to use it: rise slowly from 0 while listening to kick and bass. When the low end has body without covering the vocal, you are there. On a bus, 20–35 % is usually enough; on an instrument, 40–60 % is fair game.

TIGHT (0–100 %, default 0)

The punch. Not an EQ and not a compressor: a transient shaper that compares two envelopes of the sub band — a fast one (5 ms) and a slow one (50 ms) — and when the attack exceeds the slow one by more than 3 %, it enhances that attack up to 1.6×. **It only adds, never cuts**: it cannot squash your kick

or eat its tail.

How to use it: if the kick sounds round but doesn't punch, raise TIGHT until the hit shows up. If it already punches, leave it alone: TIGHT doesn't fix what isn't broken.

MONO BELOW (Off, 60, 80, 100, 120 Hz, default Off)

Everything below the chosen frequency goes mono. This is the tool that saves your low end in the real world: phones, small speakers, mono club systems and vinyl punish out-of-phase stereo lows — they cancel and the sub disappears. With MONO BELOW, the center stays solid while the sides keep breathing above.

How to use it: 80 Hz is a safe starting point for a bus. If your mix loses weight when summed to mono, go down to 60 Hz; if the sub still wanders side to side, go up to 100 Hz. At Off, it does nothing.

SAFETY LIMITER (On/Off, default On)

The final guardian: a limiter with a fixed −1 dBFS ceiling, placed after LOW CUT, at the very end of the chain. It is not a mastering limiter for gaining loudness — it is insurance against clips. When it isn't needed, it touches nothing; when it is, it prevents the clip without arguing.

Leave it On unless you run your own final limiter afterwards and want absolute control. The GR mode of the VU shows you exactly how much it is cutting.

AIR BYPASS (lever, default: section active)

Disconnects the whole AIR section (shelf and CLEAN) in one move. Useful for comparing brightness with and without processing, without touching the values.

AIR (0 to +6 dB, 0.1 steps, default 0)

The brightness. A gentle shelf with its corner at 8 kHz that opens the air of the mix without metal or hiss. It is the control that makes a mix sound "expensive": open, with air around vocals and cymbals.

How to use it: 1–2.5 dB already shows on a bus. If the mix starts biting as you raise it, don't lower AIR: raise CLEAN and let it negotiate.

CLEAN (0–100 %, default 0)

The policeman of the air. A bandpass detector at 4.5 kHz (Q = 1) watches

for harshness; when it finds it, it reduces **only the AIR boost, never your source**. That way you can keep the brightness high without the mix turning fatiguing.

The knob has three authority zones:

Zone	Range	Character
SOFT	0–50 %	Gentle, permissive
MID	50–83 %	Firm, negotiating
HARD	83–100 %	Strict, no mercy

On top of that, CLEAN only starts working once AIR passes roughly half its travel, with a soft knee in (no jumps). And it has a ceiling per mode: in BUS the maximum cut is about 3 dB — a bus or a master can never be torn apart —; in INST it reaches about 5.3 dB, because a single instrument has more room for correction.

How to use it: raise AIR to where you like the brightness, then raise CLEAN until the harshness calms down but the air stays alive. The DE-HARSH LED tells you when it is working.

OUTPUT GAIN (−12 to +12 dB, 0.1 steps, default 0)

The output trim. Use it to compensate for what you gained with SUB ENHANCE and AIR, and to deliver the level your next process (or your master) expects. Honest processed/unprocessed comparison happens at equal loudness: if the processed version is louder, it will always "win".

DRIVE (0–100 %, default 0)

The Vintage saturation. An asymmetric triode-like curve — even-harmonic dominant, the house "silk" — that adds body and presence without harshness. It runs at twice the sample rate (×2) so harmonics don't alias, with the wet path delay-compensated and the dry path untouched. At 0 % it is transparent: no saturation at all.

How to use it: on a bus, 5–15 % adds glue and size. On an instrument, 10–30 % gives it character. If it turns harsh, change FOCUS before lowering DRIVE: often the problem is the band, not the amount.

DRIVE FOCUS (Sub / Full / Air, default Full)

Chooses which band gets saturated:

- **Sub:** the low band only. Ideal for fattening a kick or bass without touching the rest.

- **Full:** the whole spectrum. The classic bus glue.
- **Air:** the top band only. Brightness with body, without dirtying the low end.

DRIVE POWER (Off/On, default On)

Switches the drive on and off without touching the amount knob. Leave your favorite amount dialed in and step in and out of saturation with one click.

VU METER MODE (IN / GR / OUT, default IN)

What the VU meter shows:

- **IN:** input level.
- **GR:** safety limiter gain reduction (how much it is cutting).
- **OUT:** output level.

It changes by clicking the IN/GR/OUT pilots only (the wheel does nothing here). The three pilots act as a radio selector: only one is active at a time.

POWER (Off/On, default On)

The total bypass of the plugin, with a 25 ms crossfade so stepping in and out never clicks. When Off, the meters fall, the VU goes dark (the needle rests at the left stop) and the gain-reduction report goes to 0. The DSP stops processing: this is a real bypass, not decoration.

OPERATION (Bus / Inst, default Bus)

The switch from section 4. It changes crossover, de-harsh threshold and drive voicing in one move. In REAPER's generic UI it does not appear as a knob (hidden on purpose): it changes from the GUI lever.

LOW CUT (Off, 20–200 Hz in 20 Hz steps, default Off)

The end-of-chain high-pass filter, with its green-pointer knob between SUB ENHANCE and TIGHT. Eleven positions: OFF and 20 to 200 Hz.

It sits **after the sum and before the safety limiter**. That order is the key: by removing infrasonics and DC — including any the DRIVE itself may generate — the limiter stops reacting to inaudible garbage and works on music only. The result: a cleaner low end and a limiter that breathes better, without sounding "cut".

The slope follows each mode's standard: **BUS 12 dB/oct** (2nd-order Butterworth) and **INST 18 dB/oct** (3rd-order Butterworth). The structure is SVF/TPT with coefficient smoothing (~15 ms) and an engage crossfade

(~10 ms): changing steps produces no clicks and no zipper noise. At OFF, the bypass is exact, bit for bit.

How to use it: on a bus, 20–30 Hz cleans what is inaudible without touching the music. On an instrument with handling noise or rumble, raise it until the problem disappears and the lowest note stays intact.

6. The VU meter and the LEDs

The VU is a 218 × 218 px face with a −20 to +3 dB scale, drawn from the house-approved face. Its level followers hold the true peak (150 ms release) and the needle has asymmetric smoothing (fast attack 0.75, smooth release 0.35) for a real analog-meter feel.

- **IN / GR / OUT pilots:** select what the VU shows (see above).
- **DE-HARSH LED:** signals when the harshness detector is active. Its law is deliberately eager: it lights early and climbs fast with detector activity and the CLEAN knob position. It is a warning, not an alarm: if it blinks now and then, the system is doing its job. If it lives lit, lower AIR or raise CLEAN.

7. Mouse wheel and gestures

Every knob and lever answers the mouse wheel: **120 wheel units = one step**. Equivalences per control:

Control	One wheel step
% knobs (SUB, TIGHT, CLEAN, DRIVE)	2 points
INPUT and OUTPUT	0.5 dB
AIR	0.2 dB
MONO BELOW	1 position
LOW CUT	1 step (20 Hz)
Levers (bypasses, FOCUS, OPERATION, POWER)	Directional: scrolling up/down drives them to their ends and holds them there
IN/GR/OUT selector	Click only

8. Starting recipes

These are starting points, not rules. Your ears rule; adjust from here.

Mix bus / master

OPERATION on BUS · SUB ENHANCE 20–35 % · TIGHT 15–30 % · MONO BELOW 80 Hz · AIR 1–2.5 dB · CLEAN 40–60 % · LOW CUT 20–30 Hz · SAFETY LIMITER On.

Calibrate INPUT so the VU hovers around 0 on strong hits.

Kick or bass

OPERATION on INST · SUB ENHANCE 40–60 % · TIGHT 30–50 % · MONO BELOW 60–100 Hz

· DRIVE 10–25 % with FOCUS on Sub. If the kick loses definition, lower SUB ENHANCE and raise TIGHT: punch doesn't live in low-end volume, it lives in the attack.

Harsh mix that bites

Moderate AIR (1–2 dB) + CLEAN 60–85 %. Watch the DE-HARSH LED: if it blinks a lot, the problem is the mix, not the plugin — lower AIR a little or raise CLEAN. If the LED lives lit, check the source: no de-harsh fixes a broken recording.

9. Technical specifications

Format	JSFX for REAPER (Windows, macOS, Linux)
Controls	17 (including OPERATION and LOW CUT)
Internal processing	64-bit (EEL2 doubles) throughout
Sample rates	44.1 to 96 kHz; all time constants are sample-rate independent (same behavior at any rate)
Sub crossover	4th-order Linkwitz-Riley (BUS 160 Hz, INST 90 Hz)
Sub shelf	One pole at 55 Hz, 0 to +7.5 dB
Air shelf	Corner at 8 kHz, 0 to +6 dB
De-harsh detector	Bandpass 4.5 kHz, Q = 1
Saturation	Asymmetric rational clipper, even-harmonic dominant, ×2 oversampled, wet path delay-compensated
LOW CUT	Butterworth SVF/TPT; BUS 12 dB/oct, INST 18 dB/oct; ~15 ms slew, ~10 ms crossfade; OFF = exact bypass
Limiter	Fixed −1 dBFS ceiling, last in the chain
Bypass	POWER with 25 ms crossfade
Interface	951 × 623 px metal plate, measured 1:1 from the author's layered design; English labels
Meter	218 × 218 px VU face, IN/GR/OUT modes, true peak-hold followers, 150 ms release

Design notes (for the curious): shelves are built as dry + g·filtered blends with one-pole filters — no per-sample coefficient updates, no zipper noise, trivial cost. Biquads are used only where steepness matters (sub extraction, mono lowpass, detector bandpass) and their coefficients are computed in @slider, never in @sample. Input and output are sanitized against NaN/Inf

and denormals are flushed periodically.

10. Installation

1. Unzip the package.
2. Copy `fma-sub-air.jsfx` and the `fma-sub-air_gfx` folder (with its 14 graphics) into REAPER's Effects folder (Options → Show REAPER resource path → Effects).
1. Restart REAPER or rescan FX. The plugin shows up as **FMA Sub-Air** in the FX browser.
1. Do not rename the `.jsfx` file: the graphics are located by its name.

The public version is also available via ReaPack from the FMA Series repository. This ZIP distribution is the author's protected build.

11. License and authorized use

© 2026 Free Mix Audio. All rights reserved.

- **Authorized use:** using the plugin as-is in your projects, sessions and mixes, with no track or project limits.
- **Not authorized:** modifying the code, in whole or in part; redistributing, re-publishing, uploading or sharing the plugin or its graphics without prior notice and written authorization from Free Mix Audio.
- **No implied guarantee of results:** the plugin is a tool; the final sound depends on your decisions and your material.

The full binding terms are in the `LICENSE.txt` file of this package. If you want to share it — for example, in a group or community — tell the author first: sharing with permission is welcome; sharing without permission is not.

FMA Sub-Air v1.7.1 · FMA Series · Free Mix Audio · 2026

Built so the low end weighs, punches and stays put — and so the air shines without biting.